Summary in Graph

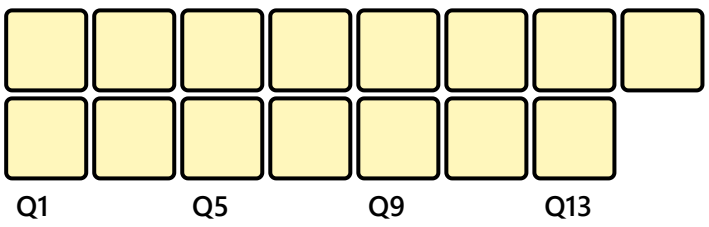
Exam Summary (GO Classes GATE CS| Relation, Lattice & Poset | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (26) out of 257 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Set Theory & Algebra

Which of the following is false?

A. Complements are unique in a complemented lattice.
B. There are no totally ordered Boolean lattices with eight elements.
C. There are no Boolean lattices with six elements.
D. There is a Boolean lattice with eight elements.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Which of the following is true?

A. Any elements x, y in a poset can be compared: either $x \leq y$ or $y \leq x$.
B. A lattice with eight elements is a boolean lattice.
C. If M is the maximum of a set S, then M is a maximal element.
D. Every total order is an equivalence relation.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

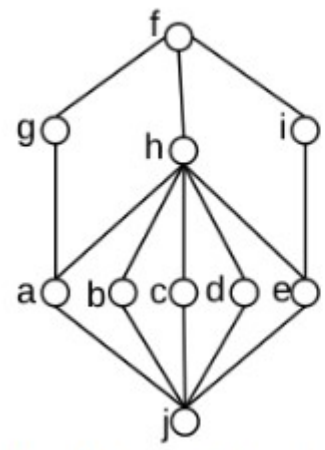
Q #3 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Let X be a finite set with 5 elements. The number of total orders on X is ?

Your Answer: | Correct Answer: 120 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Consider the following lattice



Find the number of complements for the element 'a'.

A. 3
B. 1
C. 2
D. 5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 08sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let L be a lattice. Then for every a and b in L which one of the following is correct?

A. $a \vee b = a \wedge b$

B. $a \vee (b \vee c) = (a \vee b) \vee c$

C. $a \vee (b \wedge c) = a$

D. $a \vee (b \vee c) = b$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Suppose ($A, \leq$) is a poset such that every nonempty subset of A has a least element. Which of the following is/are necessarily true for $(\mathrm{A}, \leq)$?

A. $\leq$ is a total ordering on A.
B. $(A, \leq)$ is a lattice.
C. There exists a greatest element in ($\mathrm{A}, \leq$).
D. $(\mathrm{A}, \leq)$ is a distributive lattice.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Suppose $(A, \leq)$ is a finite nonempty poset. Which of the following is/are true ?

A. A has a minimal element.
B. A has a maximal element.
C. A has a minimum element.
D. A has a maximum element.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let $(\mathrm{A}, \leq)$ be a lattice. Which of the following properties necessarily hold true for any $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathrm{A}$?

A. $x \wedge y \leq \{x, y\} \leq x \vee y$.
B. $x \leq y$ if and only if $x \wedge y = x$.
C. $x \leq y$ if and only if $x \vee y = y$.
D. $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let $(\mathrm{A}, \leq)$ be a lattice. Which of the following properties necessarily hold true for any $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathrm{A}$?

A. $x \wedge y = y \wedge x$
B. $(x \vee y) \vee z = x \vee (y \vee z)$
C. $x \wedge (x \vee y) = x$
D. $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

In a Boolean lattice, if $b \wedge \bar{c} = 0$, then

A. $\mathrm{b} \leq \mathrm{c}$.
B. $c \leq b$.
C. b,c are incomparable.
D. $\mathrm{b} = \mathrm{c}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

In a lattice ($\mathrm{A}, \leq$), an element " u " is called the dominator for join operation iff $\mathrm{x} \vee \mathrm{u} = \mathrm{u}$, for all $\mathrm{x} \in \mathrm{A}$.
In a lattice $(\mathrm{A}, \leq)$, an element " g " is called the dominator for meet operation iff $x \wedge g = g$, for all $x \in A$.
A lattice is said to satisfy Domination Law iff there exists a dominator for join operation & meet operation.
Which of the following is/are true?

A. A lattice satisfies domination law if and only if it is a bounded lattice.
B. A lattice satisfies domination law if and only if it is a complemented lattice.
C. In any lattice, the dominator for join operation is the greatest element.
D. In any lattice, the dominator for meet operation is the greatest element.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

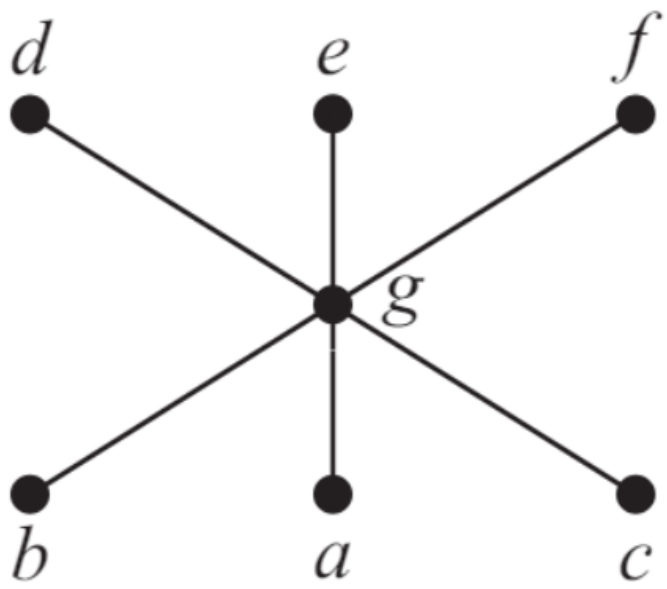
In a lattice $(A, \leq)$, an element " u " is called the identity for join operation iff $x \vee u = x$, for all $x \in A$.
In a lattice $(A, \leq)$, an element " g " is called the identity for meet operation iff $\mathrm{x} \wedge \mathrm{g} = \mathrm{x}$, for all $\mathrm{x} \in \mathrm{A}$.
A lattice is said to satisfy Identity Property iff there exists an identity element for join operation & meet operation.
Which of the following is/are true?

A. A lattice satisfies Identity Property if and only if it is a bounded lattice.
B. A lattice satisfies Identity Property if and only if it is a complemented lattice.
C. In any lattice, the identity element for join operation is the greatest element.
D. In any lattice, the identity element for meet operation is the greatest element.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

The following hasse diagram represents a POSET (A,R), where R is a partial order relation on set A.
The number of ordered pairs in the partial ordering R with the following Hasse diagram is ?? (i.e. Find the cardinality of R)



Your Answer: | Correct Answer: 22 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let T be a relation defined on Z by xTy if and only if $|\mathrm{x}| < |\mathrm{y}|$ or $\mathrm{x} = \mathrm{y}$. Which of the following is true?

A. T is Not a partial order relation.
B. T is a partial order relation, but not total order relation.
C. T is a total order relation.
D. T is an equivalence relation.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Which one of the following relations is a total order?

A. Let $A = \{a, b, c, d\}$ and let R be the relation
$R = \{(a, a), (b, b)(c, c), (d, d), (c, a), (a, d), (c, d), (b, c), (b, d), (b, a)\}$
B. $A = \{2, 3, 4\}$ ordered by the relation R on A defined as follows:
$\forall a, b \in A, aRb \longleftrightarrow (5 \bmod a) = (6 \bmod b)$
C. The set $A = \{12, 24, 48, 3, 9\}$ ordered by the divides relation.
D. None.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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